Estimating the Volatility of Wholesale Electricity Spot Prices in the US

Lester Hadsell, Achla Marathe** and Hany A. Shawky**

This paper examines the volatility of wholesale electricity prices for five US markets. Using data covering the period from May 1996 to September 2001, for the California-Oregon Border, Palo Verde, Cinergy, Entergy, and Pennsylvania-New Jersey-Maryland markets, we examine the volatility of electricity wholesale prices over time and across markets. We estimate volatility using a TARCH model to study the differences among markets and the seasonal characteristics of each market. For all markets, we find strong evidence for a downward trend in the ARCH term and a significant negative asymmetric effect over the sample period. We also document important differences among the regional electricity markets not only with respect to wholesale price volatility and seasonal variations, but also with respect to asymmetric properties and persistence of volatility.

INTRODUCTION

The recent deregulation of the electric utility industry in many parts of the United States has created competitive wholesale power markets that exhibit levels of price volatility unparalleled in traditional commodity markets.[1] The volatility of electricity prices not only varies from one year to the next, but also are predictably higher during certain times of the year – for example, during

The Energy Journal, Vol. 25, No. 4.

The authors thank Bill Ding, Chris Faugere, Adonis Yatchew (the editor), David Smith and four anonymous referees for valuable comments.

* Corresponding author: Hany Shawky, School of Business, University at Albany, Albany, New York 12222, USA. E-mail: h.shawky@albany.edu and hadsell@albany.edu

** Modeling, Algorithms and Informatics Group, Los Alamos National Laboratory, Los Alamos, NM 87545, USA. E-mail: achla@lanl.gov

1. The reason for this price behavior is often attributed to the nature of how electricity is produced and consumed, inelastic demand, uncertainty of input prices, seasonal effects and most importantly, the non-storability of electricity. These unique characteristics of the supply and demand for electricity are reflected in the behavior of wholesale power prices.

summer months when the demand for electricity is higher relative to other times of the year. Prices generally appear to be susceptible to temporary spikes and are characterized by positive skewness. Furthermore, volatility appears to be clustered and persistent.

Understanding the volatility dynamics of electricity markets is important in evaluating the deregulation experience, in forecasting future spot prices, and in pricing electricity futures and other energy derivatives. For example, Bessembinder and Lemmon (2002) point out that positive skewness in wholesale electricity prices, combined with relative fixed retail prices, may cause industry profits to decline, creating an incentive to hedge against production cost hikes by purchasing power at fixed futures prices. They assert that futures prices for electricity are a function of expected spot prices and volatility.

The literature on spot prices for electricity in the US has tended to focus on the California experience. For instance, Bailey (1998), Blumstein et al. (2002), Borenstein (2002), Borenstein, Bushnell, and Wolak (2002), Joskow (2001) and Joskow and Kahn (2002) all have investigated various issues concerning the efficiency and the performance of the California market. Other markets have been investigated by Barnett, Isaacs, and Thompson (2002), Michaels and Ellig (1999), and Klitgaard and Reddy (2000), although price volatility was not a focus. Much of the remaining empirical work on electricity markets such as Green and Newbery (1992), Wolfram (1999) and Bystrom (2000) has focused primarily on non-U.S. markets. This paper is unique in that it provides detailed estimates of volatility in electricity prices for five major US markets.

The New York Mercantile Exchange (NYMEX) began trading electricity on five regional markets in May 1996. These five markets call for electricity to be delivered at the California-Oregon Border (COB), the Palo-Verde switchyard in Arizona (PV), the South-Central U.S. (Entergy), the Midwest (Cinergy), and Pennsylvania-New Jersey-Maryland (PJM). In this paper, we use data for the period May 1996 through September 2001 to study the time series properties of spot electricity prices for these five markets and examine the regional differences and similarities in volatility.

The paper is organized in four sections. In the next section we describe the data for the five electricity markets. The data suggest annual and sub-annual variability in volatility. In section III we use a TARCH model to estimate volatility in electricity prices. We extend the basic model to capture seasonal volatility and discover market-specific seasonalities. The final section summarizes the paper and offers areas for future research.

II. DATA AND DESCRIPTIVE STATISTICS

Daily spot electricity prices for the period May 1996 to September 2001 were obtained from the New York Mercantile Exchange (NYMEX).[2] The dataset includes daily observations for all five electricity markets traded on NYMEX, where the only differences between markets are in the size of the contracts and the delivery location. Table 1 lists the five regional markets, delivery locations, service areas and the number of observations for each market. Electricity for delivery at the COB and the PV switchyard in Arizona were the first to be traded on NYMEX, beginning on the last day of May 1996. Electricity for delivery in the Midwest (Cinergy) and South-Central (Entergy) began trading two years later in June 1998. PJM began trading only in March of 1999.

Table 1. Summary of NYMEX Data Availability & Electricity Market Service Areas

Delivery Market	Spot prices begin	Spot prices end	No. of daily Obs.
COB	5/31/1996	9/7/2001	1333
Palo Verde	5/31/1996	9/7/2001	1333
Cinergy	6/12/1998	9/7/2001	818
Entergy	6/12/1998	9/7/2001	818
PJM	3/18/1999	9/7/2001	561
Delivery Market	**Delivery Location**	**Interconnection**	**Service Area**
California-Oregon Border	California-Oregon border	Western U.S.	California, Oregon, Nevada
Palo Verde	Palo Verde switchyard, AZ	Western U.S.	Arizona, California
Cinergy	Any interface into Cinergy transmission system	Eastern U.S.	Ohio, Indiana, Kentucky
Entergy	Any interface into Entergy transmission system	Eastern U.S.	Louisiana, Arkansas, Mississippi, Texas
Pennsylvania-New Jersey-Maryland	PJM western hub	Eastern U.S.	Pennsylvania, New Jersey, Maryland

2. Data obtained from NYMEX denote these daily quotes as "cash prices". These cash prices are the daily average "floating prices" for the 16 peak hours of each day, where peak hours are from 7 AM to 11 PM. Thus the word cash prices and spot prices in this paper are synonymous.

Table 2 provides sample statistics on spot prices by market and by year. A couple of aspects are worth noting. First, there is tremendous volatility in all markets. COB and PV were fairly placid in 1996 and 1997, as they were not fully deregulated until the spring of 1998 (Blumstein, Friedman, and Green, 2002). Prices and price volatility rose in 1998 and 1999 and then, as widely reported, spiked in 2000 and 2001. The average price per KWh reached $185 in the COB market for 2001, compared to under $19 in 1996. While price spikes in the California market have garnered the most attention, such spikes are not uncommon in other markets. The maximum price per KWh in the Cinergy market reached $1800 in 1998 and $1525 in 1999 (years of relative calm in California) before settling down in 2000 and 2001. For the latter years, Cinergy and Entergy experienced price volatility similar to COB and PV.

Second, significant differences are apparent between the average wholesale electricity prices among the five markets. While both COB and PV had lower prices than the other three markets in 1998 and 1999, they had substantially higher average prices in 2000 and 2001.[3] Entergy followed a similar pattern. For the three eastern markets however, average prices were fairly stable year to year throughout the study period.

Figures 1a, b and c, depict the behavior of price volatility for the five electricity markets. Figure 1a shows price volatility for COB and PV for the period from May 1996 through September 2001. Due to the unusual character of electricity prices in those markets during the later part of 2000 and the first half of 2001, the extreme volatility during that period overwhelmed the remainder of the period. To depict this behavior, we show in Figure 1b, the price volatility for May 1996 through April 2000. It is evident from Figure 1b that the COB and PV markets experienced significant volatility even prior to the events of 2000 in which price volatility reached extreme levels. Figure 1c, depicts price volatility for Cinergy, Entergy and PJM.[4] Note that while the level of price volatility in these markets is similar to the COB market, the extreme volatility spikes occur at very different time periods. In short, these observations suggest significant price volatility, regardless of region, time differences or stage of deregulation.

3. The reasons for the higher prices in COB and PV include, inter alia, higher prices for natural gas, a sharp rise in the price of tradable pollution permits, and possible market manipulation. A rise in the price of natural gas to $20 per M Btu, nearly ten times its prior price, by December 2000, led to an increase in the marginal cost of $140 per MWh for the average gas fired generator (a major source in COB and PV). Tradable pollution permits, required of generators in California to cover their emissions, increased in price in 2000, by nearly a factor of 10 (Blumstein et al, 2002, p. 24). Joskow (2001) notes, "By September 2000, Nox permit prices increased marginal supply costs from a gas-fired steam unit [in California] by $30 to $40 per MWh and increased the marginal supply costs from a peaking turbine by $100 to $120 per MWh." Investigations by federal energy regulators of possible manipulation of electricity prices in California by Enron, Avista, and El Pase Electric continues. Blumstein, Friedman, and Green (2002) provide an excellent review of the California energy crisis which began in the mid-2000 and extended into 2001.

4. While we do not show it in a separate figure, these three markets exhibited volatility patterns similar to that shown in Figure 1b.

Table 2. Summary Statistics for Spot Electricity Prices ($ kWh), NYMEX

Year(s)		COB	PV	Cinergy	PJM	Entergy
1996-2001	Mean	67.16	60.39	44.22	43.26	45.99
	Median	28.25	29.75	26.63	35.00	32.50
	St. Dev.	124.02	74.58	116.35	37.25	93.18
	Max.	3000.00	503.50	1800.00	425.00	1550.00
	Min.	7.50	9.25	10.00	16.50	14.25
	Skewness	11.18	2.67	11.22	5.84	11.71
1996	Mean	18.71	21.95			
	Median	17.50	21.00			
	St. Dev.	5.94	5.09			
	Max.	50.00	47.00			
	Min.	10.00	12.00			
	Skewness	1.32	1.54			
1997	Mean	18.49	25.52			
	Median	17.37	23.00			
	St. Dev.	5.58	10.88			
	Max.	42.12	86.21			
	Min.	8.97	10.25			
	Skewness	1.01	1.95			
1998	Mean	27.06	28.60	67.39		55:35
	Median	25.00	25.75	23.63		24.25
	St. Dev.	12.86	12.12	207.44		143.59
	Max.	82.00	86.00	1800.00		1300.00
	Min.	7.50	9.25	10.00		14.25
	Skewness	1.81	2.29	6.38		6.94
1999	Mean	30.88	31.40	45.70	41.40	45.56
	Median	29.25	30.25	21.75	26.75	23.13
	St. Dev.	10.71	9.35	139.31	55.29	127.65
	Max.	72.50	75.00	1525.00	425.00	1550.00
	Min.	15.00	18.25	12.75	16.50	15.25
	Skewness	1.16	1.54	8.96	4.57	9.23
2000	Mean	139.86	117.25	34.51	43.75	45.30
	Median	94.38	80.50	29.63	40.00	41.50
	St. Dev.	220.66	100.06	17.35	18.26	20.18
	Max.	3000.00	503.50	120.00	164.25	141.75
	Min.	26.50	24.00	14.00	21.50	16.50
	Skewness	9.09	1.52	1.82	2.40	1.44
2001	Mean	185.11	149.92	37.49	44.82	40.10
	Median	180.00	144.00	36.50	40.00	40.13
	St. Dev.	127.00	96.37	12.03	25.96	10.42
	Max.	475.00	450.00	85.00	270.00	90.00
	Min.	26.00	31.25	16.00	23.30	18.00
	Skewness	0.33	0.69	0.66	6.17	0.89

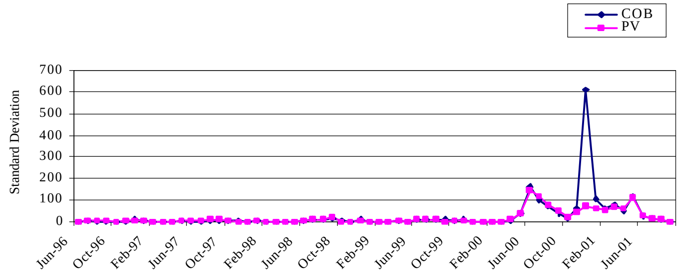

Figure 1a. Price Volatility for Electricity at the COB and PV markets, May 1996 to September 2001

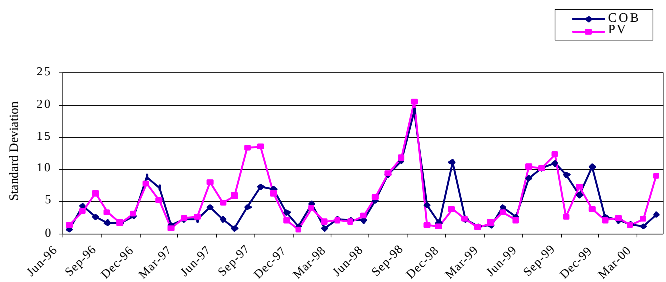

Figure 1b. Price Volatility for Electricity at the COB and PV markets, May 1996 to April 2000

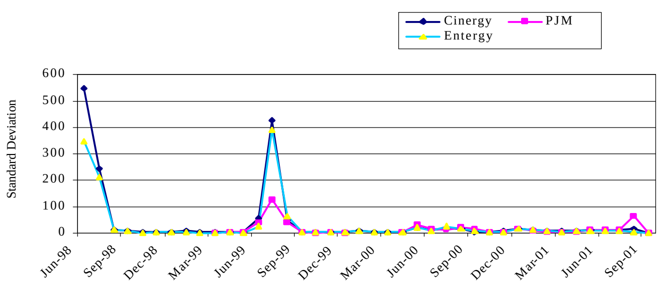

Figure 1c. Price Volatility for Electricity at the Cinergy, PJM and Entergy markets, June 1998 to September 2001

III. MODELS OF PRICE VOLATILITY WITH AND WITHOUT SEASONALITY

1. The TARCH Model

As with many other financial time series, the assumption of constant variance over time for the electricity return series is not appropriate.[5] Engle and Patton (2001) note three stylized facts about asset price volatility that are relevant to volatility in electricity markets. First, volatility exhibits persistence. Volatility is said to be persistent if today's return has a large effect on the forecasted variance many periods in the future. In other words, periods of high and low volatility tend to be clustered. Casual inspection of electricity returns suggests that this may be the case for electricity. Second, volatility tends to be mean reverting. That is, there is a normal level of volatility to which volatility eventually returns. Whether the volatility in electricity returns is mean reverting is clearly an empirical question. Third, innovations may have an asymmetric impact on volatility - positive deviations from the mean have more (or less) of an impact on volatility than negative deviations. Downward movements in equity markets, for example, are commonly followed by higher volatilities than upward movements of the same magnitude (e.g., Campbell and Hentschel, 1992; Nam, et al., 2003).

Poon and Granger (2003), in a comprehensive review of alternative methods and procedures for estimating and forecasting volatility, note the popularity of GARCH-type models for accounting for these stylized facts. While the GARCH specification of Bollershev (1986) captures the conditional variance in the spot returns, periods of high volatility followed by extended periods of relative calm suggest an asymmetric response in electricity prices that is similar to that found in other financial data. The TARCH, or Threshold ARCH, specification developed and applied by Zakoian (1994) and Glosten, Jagannathan, and Runkle (1993) is one model that handles this type of asymmetry. It also permits estimation of both persistence and mean reversion. Specifically, we estimate the following model:

$$R_t = \mu + \varepsilon_t$$

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \gamma\varepsilon^2_{t-1}\, d_{t-1} + \beta\sigma^2_{t-1}$$

The mean equation expresses spot returns as a random walk process with ε_t being the error term and the conditional variance σ^2_t, is specified as a

5. Studying the behavior of electricity prices, Bystrom (2000) and Shawky, Marathe and Barrett (2003) use a GARCH specification of Bollershev (1986) to control for changes in the variance. In such a model, the variance is "conditioned" on prior error terms, thus it allows the variance to change over time.

function of three terms: the mean ω, the news about volatility from the previous period ε^2_{t-1} (the ARCH term), and the previous period's forecast variance σ^2_{t-1} (the GARCH term).[6]

The asymmetry in the TARCH model is specified using the parameter $d_t = 1$ if $\varepsilon_t < 0$, and $d_t = 0$ otherwise. In this specification, positive errors, referred to "good news" in the empirical work on equities, and negative errors ("bad news") are expected to have differential effects on the conditional variance—"good news" has an impact of α, while "bad news" has an impact of $\alpha + \gamma$. If $\gamma > 0$, it is said that a leverage effect exists and when $\gamma \neq 0$, the news impact is asymmetric. If the behavior of electricity prices is like other financial time series, then our expectation is that γ will be positive, indicating that the occurrence of a negative return will increase volatility more than a positive return of the same size.

Black's (1976) explanation for asymmetric effects found in stock prices was that a price fall reduces the value of equity and hence increases the debt-to-equity ratio, which increases the riskiness of the firm. Black's assertion found some support by Christie (1982), Cheung and Ng (1992) and Duffee (1995). An alternative explanation uses the notion of "volatility feedback" which suggests that positive shocks to volatility increases future risk premia, which in turn, drives down current prices. This view has received some support from French et al. (1987) and Campbell and Hentschel (1992).

The point estimate of persistence, the time taken for volatility to move halfway back to its unconditional mean following a given deviation, is defined by the term $(\alpha + \frac{1}{2}\gamma + \beta)$ in the TARCH model. A value less than one implies a mean reverting conditional volatility process in which shocks are transitory in nature. With a persistence measure of less than one, the point estimate of half-life (j) in days is $(\alpha + \frac{1}{2}\gamma + \beta)^j = \frac{1}{2}$.

2. TARCH Estimates for Electricity Markets

Table 3 reports the results of the TARCH model for each of the five markets. To distinguish the period prior to full deregulation in western markets from subsequent periods, estimates were obtained for the 1996-1997, 1998-1999 and 2000-2001 sample periods for COB and PV. The first sample period captures the market before deregulation, while the latter periods depict the market after deregulation. We also estimate these parameters for each of the years and for all markets. With these delineations we can measure the components of volatility as markets evolved. After all, because of the newness of electricity markets, there is reason to suspect that market participants may require time to learn how to respond to market forces.

6. Lagged values of R_t were added to the mean equation to correct for autocorrelation as identified by inspection of the Ljung-Box Q-statistics. This correction had, generally, minor effect on the ARCH and GARCH terms.

Table 3. TARCH Model Estimation of Volatility

$$R_t = \mu + \varepsilon_t$$

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \gamma\varepsilon^2_{t-1}\, d_{t-1} + \beta\sigma^2_{t-1}$$

Year (s)		COB	PV	Cinergy	PJM	Entergy
All available days	ω	0.0006***	0.0004***	0.0018***	0.0022***	0.0012***
	(z-statistic)	(7.24)	(5.80)	(6.89)	(3.40)	(7.70)
	α	0.1863***	0.2588***	0.2565***	0.3041***	0.3749***
		(11.39)	(10.14)	(9.39)	(4.99)	(9.71)
	γ	-0.1343***	-0.1653***	-0.2343***	-0.3191***	-0.3519***
		(-7.84)	(-5.78)	(-7.57)	(-4.55)	-8.397174
	β	0.8761***	0.8319***	0.8361***	0.8151***	0.7940***
		(132.1)	(67.9)	(73.9)	(26.5)	(65.5)
	α+β	1.062	1.091	1.093	1.119	1.169
	α+½γ+β	0.995	1.008	0.975	0.960	0.993
	n	1333	1326	809	361	813
	AIC	-0.925	-1.050	-0.139	-0.139	-0.811
	DW	1.950	1.775	1.797	1.797	1.807
1996-97	ω	0.0005***	0.0016***			
		(3.78)	(4.55)			
	α	0.2968***	0.5579***			
		(6.30)	(5.86)			
	γ	-0.1333**	-0.3250***			
		(-2.22)	(-3.11)			
	β	0.8005***	0.6212***			
		(37.7)	(14.2)			
	α+β	1.097	1.179			
	α+½γ+β	1.031	1.017			
	AIC	-1.440	-1.194			
	DW	1.944	2.050			
1998-99	ω	0.0025***	0.0005***	0.0017***		0.0008***
		(5.69)	(3.18)	(5.38)		(4.72)
	α	0.4224***	0.2734***	0.3350***		0.4364***
		(3.99)	(4.86)	(6.08)		(6.34)
	γ	-0.0751	-0.1445	-0.2731***		-0.4240***
		(-0.76)	(-2.23)	(-4.54)		(-5.65)
	β	0.5793***	0.7841***	0.7997***		0.7866***
		(11.1)	(26.4)	(44.3)		(54.0)
	α+β	1.002	1.058	1.135		1.223
	α+½γ+β	0.964	0.985	0.998		1.011
	AIC	-1.115	-1.481	0.038		-0.575
	DW	2.212	1.734	1.739		1.773
2000-01	ω	0.0002***	0.0004***	0.0016***		0.0015***
		(1.30)	(3.15)	(2.22)		(3.08)
	α	0.2167***	0.2407***	0.2490***		0.3434***
		(7.98)	(5.28)	(5.28)		(6.53)
	γ	-0.2071***	-0.1741***	-0.2417***		-0.3068***
		(-7.42)	(-3.06)	(-4.22)		(-5.65)
	β	0.9106***	0.8633***	0.8543***		0.7756***
		(110.5)	(49.0)	(29.4)		(18.6)
	α+β	1.127	1.104	1.103		1.119
	α+½γ+β	1.024	1.017	0.982		0.966
	AIC	-0.361	-0.377	-0.227		-1.010
	DW	1.824	1.738	1.943		2.066

***Indicates coefficient significant at the 1% level.

Table 3. (Continued) TARCH Model Estimation of Volatility

Year		COB	PV	Cinergy	PJM	Entergy
1996	ω	0.0003*	0.0027***			
	(z-statistic)	(2.05)	(4.32)			
	α	0.8566***	0.8467***			
		(3.93)	(4.04)			
	γ	-0.4458	-0.6231**			
		(-1.58)	(-2.69)			
	β	0.6087***	0.4213***			
		(8.93)	(5.81)			
1997	ω	0.0015**	0.0006**			
		(2.66)	(2.35)			
	α	0.5455**	0.4580***			
		(2.74)	(3.48)			
	γ	-0.1538	-0.1398			
		(-1.01)	(-0.99)			
	β	0.6027***	0.6805***			
		(7.64)	(13.3)			
1998	ω	0.0023****	0.0004**	0.002		0.0013**
		(3.36)	(2.24)	(1.13)		(2.66)
	α	0.5574***	0.3195**	0.4762**		0.3138**
		(3.23)	(2.83)	(2.45)		(2.80)
	γ	0.1088	-0.0746	0.1811		-0.3445**
		(0.52)	(-0.56)	(0.75)		(-2.63)
	β	0.4297***	0.7340***	0.5646***		0.8446***
		(5.81)	(12.4)	(8.14)		(20.2)
1999	ω	0.0004**	0.0002	0.0025***	0.0006	0.0010***
		(2.54)	(1.61)	(5.34)	(1.44)	(5.78)
	α	0.0948***	0.1593***	0.4721***	0.3528**	0.6507***
		(3.60)	(3.84)	(5.37)	(3.77)	(7.57)
	γ	-0.1163**	-0.1703***	-0.5358***	-0.2437**	-0.7740***
		(-2.96)	(-3.38)	(-6.11)	(-2.39)	(-8.41)
	β	0.9431***	0.9100***	0.7928***	0.8089***	0.7767***
		(81.8)	(32.8)	(8.14)	(25.2)	(45.8)
2000	ω	0.0001	0.0003**	0.0009	0.0016**	0.0013**
		(1.20)	(2.73)	(1.98)	(2.24)	(2.85)
	α	0.1618***	0.2396***	0.2919***	0.3962***	0.4946***
		(9.54)	(4.09)	(5.24)	(3.78)	(6.08)
	γ	-0.2813***	-0.0826	-0.3270***	-0.3962***	-0.4479***
		(-12.3)	(-1.02)	(-4.58)	(-3.21)	(-5.40)
	β	0.9747***	0.8281***	0.8812***	0.8102***	0.7567***
		(167.7)	(40.5)	(33.9)	(18.3)	(18.6)
2001	ω	0.0041	0.0012	0.0105	0.0014	0.0011
		(1.38)	(1.03)	(0.83)	(1.45)	(1.04)
	α	0.2278	0.1866**	0.1008	0.1251**	0.0135
		(1.86)	(2.08)	(0.92)	(3.07)	(0.19)
	γ	-0.1600	-0.1816*	0.0044	-0.2195***	0.0148
		(-1.27)	(-2.09)	(0.02)	(-4.12)	(0.15)
	β	0.7716***	0.8980***	0.6358	0.9370***	0.9029***
		(6.62)	(17.4)	(1.70)	(20.4)	(10.7)

***Indicates coefficient significant at the 1% level.

The results in Table 3 show that for all five markets and for each period, the ARCH and GARCH terms are statistically significant at the 1% level. Furthermore, in all of five electricity markets, we observe a clear downward trend in α, the ARCH term, which measures the impact of news about volatility from the previous period. In the western markets, α was very high in both 1996 and 1997, as these markets were preparing for deregulation. The typical value for α is 0.10 to 0.20 in previous empirical work on other commodities and equities.[7] In the case of COB, α was 0.86 in 1996, and was 0.54 in 1997. For the year 1998, COB's first year of full deregulation, α remained high at a value of 0.56. Once full deregulation took effect, the value for α began to quickly decline to 0.09, 0.16, and 0.23 for the following three years (1999 to 2001). A similar pattern held for PV and for the three eastern markets.

We offer two explanations for the decline in α values. First, in the cases of COB and PV, where α's are the highest, the α might reflect the effects of a bumpy deregulation process. A second explanation that is consistent with the first, is that in all five markets, participants were learning how electricity markets worked, basing decisions more on prior day returns than on long-run expectations. After all, forming long-run expectations of changing prices were new to markets participants, that until recently were regulated.

A high value for α implies an unstable expected volatility. The early years of partial market deregulation may have been characterized by slow price adjustments. A high α also implies an exaggerated reaction by market participants to prior day return errors, ε^2. This behavior is likely to generate price extremes. Unsure of what to expect, traders may have "played it safe," adjusting prices to normal levels only slowly. Either way, "new market syndrome," seems an especially appealing explanation since the same pattern was evident in all five markets. Moreover, the steady decline in α is matched in all markets by a visible, although less consistent increase in β, which measures the impact of the forecasted variance from last period. This indicates a movement of volatility dependence from new information to past volatility.

The asymmetric effect, captured by γ, is negative and statistically significant for the entire sample period for all five markets. The negative coefficients indicate a strong market response to "negative" news ($\varepsilon_t < 0$), in a direction that is counter to what is found in other financial time series. In the case of COB and PV, this asymmetric effect was not statistically significant in individual years before full deregulation, but was very evident in the first three-year period (1996-1998) and the three following years (1999-2001). For this latter period, the estimates for the γ coefficient range between -0.08 and -0.28 for these two markets, which is close in absolute terms to the values for α. Thus, while "good" news ($\varepsilon_{t-1} > 0$) appears to have a positive effect on conditional volatility,

7. This was also the case for the results obtained for COB and PV for the three-year samples 1996-1998 and 1999-2001 (not reported here).

"bad" news ($\varepsilon_{t-1} < 0$) has, on average, little effect. The asymmetric effect was strong and negative in the Cinergy, Entergy, and PJM markets until 2001. The effect was quite high in absolute terms, often exceeding the ARCH effect (value of α).

Our explanation for this interesting anomaly lies in the definition of "good" and "bad" news. For equities, "good" means $\varepsilon_t > 0$, but for electricity "good" means $\varepsilon_t < 0$. That is, returns that exceed the mean are desirable for equities, but are not for electricity. Given inelastic demand and the non-storability of electricity, when electricity prices exceed the mean, traders may become panicked, as they must purchase electricity at a much higher price. Thus, prices above the mean lead to an asymmetric response.[8] With these definitions of "good" and "bad" news pertaining to electricity, our findings parallel those for equity.

We also find regional differences in persistence of volatility. Volatility over the entire sample period was quite persistent in the western markets, less so in Entergy, and quite transitory in Cinergy and PJM. In the western markets before deregulation, the statistically significant persistence estimates were generally greater than one. After deregulation, persistence estimates fell below one. For the entire sample period, the persistence estimate for COB was not significantly different from 1 (F-stat = 0.561). Nevertheless, with a value of 0.995, the half-life is calculated to be 144 days. For Palo Verde, the value for ($\alpha + \frac{1}{2}\gamma + \beta$) is not statistically different from 1 (F-stat = 0.976), and with a value of 1.008: its half-life cannot be calculated, suggesting that shocks are permanent.

The half-life is much lower for Cinergy, at 28 days. And while the persistence estimate is not significantly different from 1 (F-stat = 2.635), the half-life for PJM is 17 days and 98 days for Entergy. Most significantly, by year 2001, all markets except PV (113 days) had very short half-lives: 8 days for COB, 2 days for Cinergy, 14 days for PJM, and 9 days for Entergy.

In the TARCH model it is possible to separately measure the decay following positive and negative shocks. Positive shocks decay at ($\alpha + \beta$), so the estimate of half-life in days is $(\alpha + \beta)^j = \frac{1}{2}$. Only in the case of Cinergy and Entergy during the year 2001, and COB in year 1998, did positive shocks exhibit decay. In all other periods and for all five markets, this persistence measure was greater than one. In other words, positive shocks in electricity markets exhibited a permanent impact, indicating that returns exceeding the mean led to an increase in conditional volatility that did not die down. In short, the effects of a negative shock dissipated much more quickly than the effects from a positive shock.

3. TARCH Model with Seasonal Dummies

Given the unique supply and demand characteristics of electricity (Borenstein, 2002; Michaels and Ellig, 1999), unique seasonal patterns might

8. See Michaels and Ellig (1999) and Borenstein (2002) for a discussion of supply and demand factors in electricity markets.

be expected in electricity volatility. To examine the potential seasonal components of the level and volatility of electricity prices, we re-estimate the TARCH model incorporating eleven seasonal dummies to represent months. The specification for the conditional variance with seasonal dummies is now represented as:

$$R_t = \mu + \varepsilon_t$$

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \gamma\varepsilon^2_{t-1} d_{t-1} + \beta\sigma^2_{t-1} + \Sigma\phi_i D_i$$

where $D_1 = 1$ for January spot prices and 0 otherwise, $D_2 = 1$ for March spot prices and 0 otherwise, and so on (February, the most placid month, is the base).

The results for the overall time period are presented in Table 4. An asterisk indicates statistically significant coefficients. The results shown in Table 4 are consistent with those found earlier in Table 3 except for the additional coefficients for the dummy variables. Unlike the results of Table 3, the constant in the variance equation is not statistically significant in three markets, implying an important seasonal effect in the volatility estimation. Moreover, for all five markets, the ARCH and GARCH terms are statistically significant at the 1% level.

The estimates for α, β, and γ are changed little when seasonal dummies are added.[9] The estimated coefficients for the seasonal dummies in Table 4 reveal important differences in the volatility between the five electricity markets. For the COB and the PV markets, and to a lesser extent the Entergy market, there does not seem to be any particular month in which a seasonal spike in volatility is apparent. Alternatively, we observe a significant seasonal coefficient in these three markets for almost every month of the year. Moreover, the magnitude of the coefficient on the dummies have larger values in some months, such as May and August for COB and PV, December for COB and June for PV. This suggests that while each month exhibits different volatility behavior, none is distinctively unique with respect to its magnitude or its volatility.

9. The half-life of volatility shocks is 11 days for COB, 4620 days for PV, 24 days for Cinergy, 165 days for PJM, 165 days, and 28 days for Entergy.

Table 4. TARCH - Seasonal Variation in Conditional Volatility

$$R_t = \mu + \varepsilon_t$$

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \gamma\varepsilon^2_{t-1} d_{t-1} + \beta\sigma^2_{t-1} + \Sigma\phi_i D_i$$

(z-stats in parentheses)

All Data	COB	PV	Cinergy	PJM	Entergy
ω	-0.0002***	-0.0002***	0.0012*	0.0001	0.0001
	(-2.94)	(-11.1)	(1.91)	(0.14)	(1.09)
α	0.1693***	0.2564***	0.2329***	0.1868***	0.3076***
	(9.98)	(9.11)	(7.12)	(5.64)	(8.72)
γ	-0.1053***	-0.1592***	-0.2342***	-0.3004***	-0.3393***
	(-4.89)	(-4.57)	(-6.21)	(-6.51)	(-8.21)
β	0.8657***	0.8206***	0.8423***	0.9422***	0.8177***
	(81.1)	(61.7)	(50.9)	(50.1)	(57.6)
January	0.0001	0.0007***	0.0009	0.0021	0.0007
	(0.08)	(2.34)	(0.98)	(-0.77)	(1.83)
March	0.0004***	0.0008***	-0.0008	-0.0011	0.0003
	(3.33)	(5.15)	(-1.04)	(-1.17)	(1.54)
April	0.0005***	0.0007***	0.0007	-0.0011	0.0015***
	(2.95)	(4.72)	(0.89)	(1.60)	(3.73)
May	0.0041***	0.0025***	0.0021	0.0005	0.0023***
	(6.29)	(3.18)	(1.63)	(0.72)	(3.65)
June	0.0009**	0.0022***	0.0134***	0.0032**	0.0051***
	(2.01)	(2.95)	(4.16)	(2.31)	(4.87)
July	0.0002	0.0009	0.0007	0.0011	0.0008
	(0.36)	(1.22)	(0.18)	(0.73)	(0.69)
August	0.0016***	0.0017***	0.0006	-0.0003	0.0026***
	(3.11)	(2.42)	(0.36)	(-0.36)	(2.37)
September	0.0006***	-0.0001	0.0011	0.0017	0.0019***
	(4.15)	(0.06)	(0.78)	(1.36)	(2.89)
October	0.0004***	0.0005***	-0.0012	-0.0017***	0.0001
	(3.64)	(4.12)	(-1.67)	(-2.49)	(0.55)
November	0.0006***	0.0009***	0.0031***	0.0016*	0.0015***
	(3.75)	(3.67)	(3.20)	(1.92)	(3.91)
December	0.0047***	0.0003	0.0013	0.0051	0.0021***
	(11.9)	(1.76)	(1.13)	(1.59)	(3.26)
n	1327	1327	817	361	812
AIC	-1.013	-1.033	-0.154	-0.527	-0.867

The estimated coefficients for the seasonal dummies for PJM and Cinergy show a very different pattern. For PJM, only June and October exhibit a statistically significant seasonal coefficient. For Cinergy, significant seasonal coefficients are observed for the warm month of June and again for the cold month of November. For these two eastern markets, there appears to be demand-side seasonal variations in electricity price not only in summer, but also in cold weather. Moreover, the three markets in which natural gas is a primary input, COB, PV, and Entergy, exhibit greater seasonal volatility. Dramatic price increases for natural gas had proportionately greater impact on these markets than it did on the others, even for similar fluctuations in demand. While market-specific differences in supply and demand clearly seem to be important factors explaining seasonal volatility, exploring these complex relationships is outside the scope of this paper.

IV. CONCLUSION

We study the volatility of wholesale electricity prices over time and across five markets in the United States. Our data covers the period from May 1996 to September 2001, for the California-Oregon Border, Palo Verde, Cinergy, Entergy, and Pennsylvania-New Jersey-Maryland markets. We estimate a TARCH model to study price volatility and seasonal characteristics of each of the markets.

In all five markets, we find strong evidence for a downward trend in the estimated ARCH term over the sample period implying unstable expected volatility. The early years of partial market deregulation appear to have been characterized by slow price adjustments. Exaggerated reaction by market participants to prior day return errors in the early years is likely to have generated price extremes. Unsure of what to expect, traders may have "played it safe," adjusting prices to normal levels only slowly. This pattern suggests the presence of a "new market syndrome" phenomenon. The steady decline in the ARCH term is matched in all markets by a visible, although less consistent increase in the GARCH term, which measures the impact of the forecasted variance from last period, indicating a movement of volatility dependence from new information to past volatility.

We find an unexpected asymmetric effect that is negative and statistically significant for the entire sample period for all five markets. The negative coefficients indicate a strong market response to "negative" news in a direction that is counter to what is found in other financial time series. In the case of COB and PV, this asymmetric effect was not statistically significant in individual years before full deregulation, but was very evident in the years following deregulation. We explain this apparent anomaly by redefining “good” and “bad” news for electricity markets.

We also find regional differences in persistence of volatility. Volatility over the entire sample period was quite persistent in the western markets, less so in Entergy, and quite transitory in Cinergy and PJM. In the western markets before deregulation, the statistically significant point persistence estimates were generally greater than one. After deregulation, persistence estimates fell below one. For the entire sample period, the point persistence estimate for COB was not significantly different from one.

Regional differences in supply and demand characteristics, an issue beyond the scope of this paper, is likely to play a role in explaining volatility differences among electricity markets. For instance, the COB and PV markets, with high volatility, but no seasonal pattern, may primarily reflect their reliance on volatile natural gas inputs. It is important to also remember that the 2000 and 2001 period in the California market experienced unusual circumstances that had significant impact on the level and volatility of natural gas and electricity prices. Blumstein, Friedman and Green (2002) provide a detailed documentation of the California energy crisis.

The seasonal patterns of volatility in Cinergy and PJM are almost exactly opposite of those for COB and PV, June and late autumn returns are significantly higher than those in other months. Both Cinergy and PJM rely heavily on coal to generate electricity, where they sometimes experience unexpected and pronounced shifts in demand related to weather. Furthermore, because the Entergy market is similar to COB and PV in its reliance on natural gas, its volatility behavior is in some ways more similar to western markets than the eastern markets.

Because COB and PV are on the western U.S. interconnection and Cinergy, PJM, and Entergy are on the eastern interconnection, it is not likely that these markets will converge in the near future.[10] Patterns of price volatility and seasonal variations among markets presented in this study are likely to persist for a long time, especially given the nature of electricity as a non-storable commodity and the unique supply and demand conditions that prevail in each of the markets.

While we focus on estimating volatility across five markets and across time and assessing the results for a relatively short period of time, future research should reexamine the behavior of electricity prices as the markets mature and more data becomes available. Furthermore, the determinants of volatility which is likely a function of unique regional supply and demand dynamics is subject for future inquiry.

10. See Energy Information Administration (2002, p. 34). In fact, bottlenecks within an interconnection mean that prices in connected markets, such as COB and PV, may not converge. And, in fact, while COB and PV prices and volatility were found to be similar they were not the same for the period studied.

REFERENCES

Barnett, Andy H., Justin P. Isaacs, and Henry Thompson (2002). "Retail Electric Competition and Free Interstate Trade in the Southeast." *The Electricity Journal*, March: 68-74.

Bailey, Eliza M. (1998). "Electricity Markets in the Western United States." MIT Center for Energy and environmental Policy Research, no. 143.

Bessembinder, Hendrik and Michael L. Lemmon (2002). "Equilibrium Pricing and Optimal Hedging in Electricity Forward Markets." *Journal of Finance* 57(3): 1347-1382.

Black, F. (1976). "Studies of stock price volatility changes." *Proceedings of the American Statistical Association, Business and Economic Statistics Section:* 177-181.

Blumstein, Carl, L.S. Friedman, R.J. Green (2002). "The History of Electricity Restructuring in California." Center for the Study of Energy Markets, Working paper #103, August.

Bollerslev, T. (1986). "Generalized autoregressive conditional heteroskedasticity." *Journal of Econometrics* 31: 307-327.

Borenstein (2002). "The Trouble With Electricity Markets: Understanding California's Restructuring Disaster." *Journal of Economic Perspectives* 16(1): 191-211.

Borenstein, Severin, James Bushnell, and Frank Wolak (2002). "Measuring Market Inefficiencies in California's Restructured Wholesale Electricity Market." Center for the Study of Energy Markets, Working paper # 102, June.

Bystrom, Hans NE (2000). "The Hedging Performance of Electricity Futures on the Nordic Power Exchange Nord Pool." Working paper.

Campbell, John Y. and Ludger Hentschel (1992). "No News Is Good News: An Asymmetric Model of Changing Volatility in Stock Returns." *Journal of Financial Economics* 31(3): 281-318.

Cheung, Y. W. and L. K. Ng (1992). "Stock Price Dynamics and Firm Size: An Empirical Investigation." *Journal of Finance* 48: 985-997.

Christie, A. (1982). "The Stochastic Behavior of Common Stock Variances: Values, Leverage, and Interest Rate Effects." *Journal of Financial Economics* 10: 909-927.

Duffee, G. R. (1995). "Stock Returns and Volatility: A Firm Level Analysis." *Journal of Financial Economics* 37: 399-420.

Energy Information Administration (2002). "Derivatives and Risk Management in the Petroleum, Natural Gas, and Electricity Industries." October. Accessed at http://www.eia.doe.gov/oiaf/servicerpt/derivative/index.html.

Engle, Robert F. and Andrew J. Patton (2001). "What good is a volatility model?" *Quantitative Finance* 1: 237-245.

French, K., W. Schwert and R. Stambaugh (1987). "Expected stock returns and volatility." *Journal of Financial Economics* 19: 3–29.

Glosten, Lawrence R., Ravi Jagannathan and David E. Runkle (1993). "On the Relation between Expected Value and the Volatility of the Nominal Excess Returns on Stocks." *Journal of Finance* 48(5): 1779-1801.

Green, R. and D. Newbery (1992). "Competition in the British electricity spot market." *Journal of Political Economy* 100: 929-953.

Joskow, Paul and Edward Kahn (2002). "A Quantitative Analysis of Pricing Behavior In California's Wholesale Electricity Market During Summer 2000." *The Energy Journal* 23(4): 1-35.

Joskow, Paul L. (2001). "California's Electricity Crisis." *Oxford Review of Economic Policy* 17(3), Autumn: 365-388.

Klitgaard and Reddy (2000). "Lowering Electricity Prices Through Deregulation." *Current Issues in Economics and Finance, Federal Reserve Bank of New York* 6(14): 1-6.

Michaels, Robert J. and Jerry Ellig (1999). "Electricity: Price Spikes by Design?" *Regulation* 22(2): 20-22.

Nam, Kiseok, Chong Soo Pyun and Sei-Wan Kim (2003). “Is asymmetric mean-reverting pattern in stock returns systematic? Evidence from Pacific-basin markets in the short-horizon.” *International Financial Markets, Institutions, and Money* 13: 481-502.

Poon, Ser-Huang and Clive W.J. Granger (2003). “Forecasting Volatility in Financial Markets: A Review.” *Journal of Economic Literature* 41(2): 478-539.

Shawky, Hany A., Achla Marathe and Christopher L. Barrett (2003). “A First Look at the Empirical Relation between Spot and Futures Electricity Prices in the US.” *Journal of Futures Markets* 23(10): 931-955.

Wolfram C. (1999). “Measuring duopoly power in the British electricity spot market.” *American Economic Review* 89: 805-826.

Zakoian, J. (1994). “Threshold Heteroskedastic models.” *Journal of Economic Dynamic and Control* 18: 931-995.